MYSX-256/512: A 256-bit Block Cipher with 512-bit Keys (v0.2-draft)

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Abstract

We present MYSX-256/512 v0.2-draft, an experimental 256-bit block cipher with 512-bit keys and 128 rounds. The design combines Feistel cells with Simon's F function, a Myszkowski byte transposition, word rotations, and an ARX key schedule. The cipher core (round function, F, transposition, key schedule) is unchanged from v0.1-draft2.

The v0.2 revision changes two things:

1. **Rotation schedule.** The consecutive schedule `(r mod 31)+1` of v0.1 is replaced by `rand_s9_45430` (31 explicit values in §2.8), selected by systematic search over 1M random 31-period schedules. On the 9 metrics measured for v0.2 — integral at R=5/6, linear at R=10/14 both directions, exact DMITM, weight-0 chain, MILP differential at R=5 — the new schedule **improves 6, ties 3, and regresses 0**.
2. **AEAD packet v2 (VERSION=0x02).** A new namespace `v2` adds: 512-bit secrets processed through HKDF-SHA512 (eliminating the 256-bit HKDF-SHA256 bottleneck of v1), 512-bit MAC keys, HMAC-SHA512 with 64-byte tags, one key per message (32-byte salt), authenticated header, rotation-calendar selected by a header byte, and Argon2id for password paths. The v1 API (consecutive calendar, HKDF-SHA256, PBKDF2-SHA256, 32-byte tag) is kept for reading legacy packets and is marked deprecated for new data.

Measured results for v0.2:

Metric	v0.1	v0.2
Exact DMITM	17 rounds	**9**
Weight-0 chain	13	**5**
Truncated impossible	39	**28**
Integral dp5 / dp6 (r0=0, word0)	21 / 4	**9 / 4**
MILP differential R=5	0	**4**
Linear single-trail R=10 fwd / bwd	1 / 1	1 / **3**
Linear single-trail R=14 fwd / bwd	5 / 5	**8 / 11**
Linear single-trail R=128 fwd	2248 (bwd)	2294 (fwd)
v2 cross-language KAT	—	6/6 packets byte-identical
v2 security tests	—	58/58 pass

What is PENDING for v0.2 (measured for v0.1, not for v0.2): full 31-offset differential sweep (currently 6 offsets, best r0=29 weight 94, 2 offsets `None`); beam differential (MILP only for v0.2); linear backward at R>14 and R=128; hull empirical verification; integral sweep outside r0=0 (only 6 offsets); NIST SP 800-22; related-key re-validation.

Verdict. v0.2 improves 6 of 9 measured metrics, ties 3, and does not regress. The main weakness of v0.1 — a differential trail of weight 2 at R=14 for offset r0=10, giving a distinguisher at 4 chosen pairs — is eliminated in v0.2 (best MILP at R=14 is 94, with 2 offsets unresolved). v0.2 is not proven secure, has not been externally audited, and must not be used to protect real data.

Executive summary

v0.2 changes the rotation schedule and the AEAD packet format. The block cipher core is unchanged.

Rotation schedule. The consecutive schedule `(r mod 31)+1` of v0.1 gives a differential trail of **weight 2 at R=14 for offset r0=10**, a 13-round weight-0 chain, and 17-round exact DMITM. These are not "tight margins"; they are structural alignments between the transposition and the consecutive step. The v0.2 calendar `rand_s9_45430` was selected by a systematic search over 1M random 31-period schedules followed by an 8-candidate evaluation on 9 metrics. It gives a 5-round weight-0 chain, 9-round exact DMITM, 28-round truncated impossible, and a best MILP differential weight at R=14 of 94 (cota superior; 2 offsets unresolved). The v0.1 offset r0=10 that was catastrophic for the differential analysis is eliminated in v0.2.

AEAD v2. HKDF-SHA512 with a 64-byte secret restores the `2^256` Grover bound; 512-bit MAC keys and HMAC-SHA512 tags replace the 256-bit HMAC of v1; the rotation calendar is selected by header byte `CAL` instead of hard-coded or environment variable; one key per message (32-byte salt) eliminates nonce reuse; Argon2id is used for password paths with explicit bounds. A 58-test security suite verifies the API; a cross-language KAT (Python reference vs JavaScript implementation) confirms 6/6 packets byte-identical.

Not established. No lower bound over 128 rounds. No key-recovery attack. No NIST SP 800-22 for v0.2 (v0.1 passed). No external audit. The v0.2 analysis is partial by design; PENDING items are listed in §4.14 and §8.

1. Introduction

MYSX-256/512 is an experimental block cipher. It is offered as a public proposal for cryptanalysis, not as a production primitive. The author makes no claim of security beyond the observed results reported here, and explicitly welcomes external review.

1.1. Security objectives (and non-objectives)

Model. Single-key, chosen-plaintext for confidentiality, combined with an encrypt-then-MAC AEAD wrapper for integrity (§2.9). No claim under related-key, chosen-ciphertext, or side-channel models.

Claimed security level. None formally. The paper reports observed upper bounds on single-trail weights (§3), a verified exact model of F and of the round (§4.2–§4.6), beam searches over that model (§4.10), and impossible differentials in both exact and truncated form (§4.12). These are inputs to cryptanalysis, not proofs.

Reading thresholds. A differential trail of weight w needs about 2^w chosen pairs; a linear trail of weight w has |corr| = 2^-w and needs about 2^(2w) known texts. A 2^128-text budget is a differential weight of ≈128 but a **linear** weight of ≈64. Compare w_diff with 2·w_lin; never compare them directly.

Non-objectives. MYSX v0.2 does not claim:

- post-quantum security beyond Grover: 2^256 iterations for a uniformly random 512-bit secret processed through HKDF-SHA512 (2^128 for a 256-bit secret through HKDF-SHA256 in the v1 API);
- resistance to related-key attacks beyond the heuristic coverage of §5 (key schedule unchanged from v0.1);
- resistance to integral attacks beyond the R=5/6 distinguishers at r0=0 and the 6-offset sweep (§4.9);
- resistance to linear-hull effects beyond the single-trail weights of §4.10 (no hull-aware bound beyond R=17);
- constant-time execution in JavaScript (side channels out of scope).

1.2. Design rationale

MYSX combines four building blocks, unchanged from v0.1:

- **Simon's F** F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2). Chosen for its studied diffusion/linear/differential properties in Simon and its simplicity (no S-boxes, only AND/XOR/rotl). It does not inherit Simon's security proofs.
- **Feistel cells over 32-bit word pairs.** Four parallel cells per round, then a byte transposition over the full 32-byte state, then word rotations. Classified as an SPN-like structure with Feistel cells, not as a 2-branch Feistel.
- **Myszkowski transposition** (§2.3.1). A fixed byte permutation derived from the keyword MYSZKOWS. Classical transposition with keyword; provides byte-level diffusion. A permutation optimized for diffusion would strictly dominate it and is a candidate for a future version.
- **ARX key schedule** (§2.4). A bijective Feistel 8+8 update.

Rotation schedule (v0.2). The consecutive schedule (r mod 31)+1 of v0.1 is replaced by rand_s9_45430 (31 values in §2.8). The selection is systematic (§4.13). The consecutive schedule is not "moderately weak": at 3000 random schedules sampled without filtering, it is in the worst ~2% by exact DMITM and worst ~0.2% by weight-0 chain. The cause is the step of 1 aligning with the transposition period; the permutation is not the cause (40 random byte permutations with the consecutive schedule give mean exact DMITM 21.4).

Parameter choices. 256-bit block, 512-bit key, 128 rounds. Not derived from a formal argument. The round count is a parameter, not a security level.

1.3. Comparison with related designs

Design	Block	Key	Rounds	Structure	S-boxes
Simon128/128	128	128	68	Feistel	No

Design	Block	Key	Rounds	Structure	S-boxes
Speck128/256	128	256	34	ARX	No
ChaCha20	512	256	20	ARX	No
MYSX-256/512	256	512	128	SPN-like with Feistel cells	No

MYSX's closest sibling is Simon. The differences (block size, branch organization, transposition, word rotations, non-linear key schedule) are substantial enough that Simon cryptanalysis does not transfer directly.

Performance. MYSX is estimated to run at a few MB/s in JavaScript, versus hundreds of MB/s for AES-GCM via WebCrypto. MYSX is not competitive on performance; it is a research primitive.

1.4. Versioning

The reference implementation (`mysx.prod.js`) is labelled v0.2-experimental. This document is the v0.2-draft of the design. The v0.1-draft2 paper (tag `v0.1-draft2` in the repository) is the previous design; its full analysis is preserved in `LEGACY_V1/`.

2. Specification

2.1. Parameters

Parameter	Value
Block size	256 bits (8 words of 32 bits)
Key size	512 bits (16 words of 32 bits)
Rounds	128
Structure	SPN-like: 4 parallel Feistel cells + byte transposition
Non-linearity	`F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2)`
Diffusion	Myszkowski transposition (keyword MYSZKOWS)
Rotation (v0.2)	**`ROT_CAL[r mod 31]`** (§2.8)
Final whitening	XOR with 8 words from the key schedule

2.2. Notation

- `rotl(x, n)`: circular left rotation of `x` by `n` bits.
- `∧`: bitwise AND.
- `⊕`: bitwise XOR.
- `s[r]`: state at round `r` (`s[0]` = plaintext, `s[128]` = ciphertext).
- `s[r][i]`: word `i` (0..7) of the state at round `r`.
- `rk[r][i]`: subkey `i` (0..7) of round `r`.
- `rkFinal[i]`: word `i` of the final whitening.
- `ROT_CAL[]`: the 31-value rotation schedule (§2.8).
- `CAL`, `KDF`: one-byte selectors in v2 packets (§2.9).

2.3. Constants

```
CONST[0..7] = [
  0x243f6a88, 0x85a308d3, 0x13198a2e, 0x03707344,
  0xa4093822, 0x299f31d0, 0x082efa98, 0xec4e6c89
]
```

These are the fractional digits of π (same as Blowfish).

2.3.1. Myszkowski permutation Keyword: `MYSZKOWS` (8 letters, S repeated at positions 3 and 8).

```
MYZ_PERM = [
   4,12,20,28,                 # K (column 4)
   0, 8,16,24,                 # M (column 0)
   5,13,21,29,                 # O (column 5)
   2, 7,10,15,18,23,26,31,     # S,S (columns 2 and 7, row-wise)
   6,14,22,30,                 # W (column 6)
   1, 9,17,25,                 # Y (column 1)
   3,11,19,27                  # Z (column 3)
]
```

The inverse `MYZ_INV` satisfies `MYZ_INV[MYZ_PERM[i]] = i`.

2.4. Key schedule

Initialization. `s[0..15]` from 64-byte key (big-endian per word).

Expansion. For `r = 0..127`:

```
spread = (r + 1) * 0x9E3779B9  (mod 2^32)
for i in 0..7:
    rk[r][i] = s[i] ⊕ s[(i+5) mod 16] ⊕ s[(i+11) mod 16]
               ⊕ CONST[(r+i) mod 8] ⊕ spread
for i in 0..3:
    fk = s[(i+8) mod 16] ⊕ CONST[(r+i+3) mod 8]
    rk[r][i] = rk[r][i] ⊕ fk
L, R = s[0..7], s[8..15]
for i in 0..7:
    a, b, c = R[i], R[(i+1) mod 8], R[(i+3) mod 8]
    F = rotl((a + b) mod 2^32, 7) ⊕ rotl(c, 13) ⊕ CONST[i] ⊕ spread
    s[i], s[i+8] = R[i], L[i] ⊕ F
```

Final whitening. `rkFinal[i] = s[i] ⊕ s[(i+8) mod 16] ⊕ CONST[(128+i) mod 8]`.

2.5. Round function

For round `r`:

Step 1. `s[r][i] = s[r][i] ⊕ rk[r][i]` for i = 0..7.

Step 2. For p = 0..3:

`L, R = s[r][p], s[r][p+4]`
`s_mid[p], s_mid[p+4] = R, L ⊕ F(R)`

Step 3. Convert `s_mid[0..7]` to bytes (big-endian per word), apply `permuted[j] = bytes[MYZ_PERM[j]]`, reconstruct words.

Step 4. `shift = ROT_CAL[r mod 31]`; `s[r+1][i] = rotl(s_perm[i], shift)`.

2.6. Encryption

`ciphertext = MYSX_encrypt(plaintext, key, ROT_CAL)` where the last step XORs `rkFinal`.

2.7. Decryption

Inverse of §2.6 in reverse order.

2.8. Rotation schedule (v0.2)

The 31-value rotation schedule is:

```
ROT_CAL = [
  22,  5,  8,  7, 17, 10, 30,  1, 11, 26,
   6,  3, 12, 23, 25, 13, 28, 20, 21, 19,
  24,  2, 18,  9,  4, 15, 29, 27, 16, 14,
  31
]
```

Round `r` uses `shift = ROT_CAL[r mod 31]`. The v0.1 schedule was `shift = (r mod 31) + 1` (i.e., `[1,2,...,31]`).

Selection method. 1M random permutations of `0..30` filtered by:

1. Maximum weight-0 chain at offset 10 ≤ 3,
2. Global maximum weight-0 chain ≤ 5,
3. Exact DMITM over the full left-half difference space.

8 candidates achieved exact DMITM = 9. The 8 finalists were ranked by a multi-dimensional score: integral at R=5/6, linear at R=10/14 (both directions), MILP differential at R=5, exact DMITM, weight-0 chain.

Why the consecutive schedule was rejected. Over 3000 random schedules without filtering, the consecutive schedule is in the worst ~2% by exact DMITM and worst ~0.2% by weight-0 chain. The cause is the step of 1: a consecutive sequence of shifts aligns with the transposition period in a way that lets the tracked bits remain in the "safe" half of the state for many rounds. The variable is the schedule step, not the Myszkowski permutation (a control with 40 random byte permutations and the consecutive schedule gives mean exact DMITM 21.4).

The differential weakness closed by v0.2. In v0.1, at offset r0=10 the best single-trail differential weight at R=14 is **2** (i.e., 4 chosen pairs distinguish 14 rounds of v0.1 from a random cipher). In v0.2, the best MILP weight at R=14 over 31 offsets is **94**

(offset r0=29; two offsets r0=19 and r0=27 timed out and are candidates for lower values). The structural alignment is gone.

Reproducibility. `CALENDAR_v02.txt` contains the schedule as `MYSX_ROT_CAL=22,5,8,...`. `search_mysx_deep.py --n 1000000 --seed 9` regenerates the search. `cryptanalysis/DMITM/wf2_deep_summary.json` contains the raw 8×9 table.

2.9. AEAD packet format (v2)

Two packet formats coexist. v1 (VERSION=0x01) uses the consecutive calendar, HKDF-SHA256, HMAC-SHA256, 32-byte tag; deprecated for new data. v2 (VERSION=0x02) is recommended:

```
VERSION(1)=0x02 | CAL(1) | KDF(1) | KDF_PARAMS(var) | SALT(32) | NONCE(16) | CT(n) | TAG(64
```

Fields.

- `CAL`: rotation calendar selector. `0` = consecutive (v0.1-compatible), `1` = `rand_s9_45430`. Unknown values rejected.
- `KDF`: `0` = RAW (64-byte or 32-byte secret, direct HKDF), `1` = Argon2id, `2` = PBKDF2-SHA512, `3` = scrypt.
- `KDF_PARAMS`: bounds checked on parse. Argon2id `memKiB` ∈ [19456, 1048576], `passes` ∈ [2, 16], `lanes` ∈ [1, 16]; PBKDF2-SHA512 `iterations` ∈ [210000, 10000000]; scrypt `logN` ∈ [15, 20], `r` ∈ [8, 16], `p` ∈ [1, 16], memory ≤ 1 GiB.
- `SALT` (32 bytes), `NONCE` (16 bytes), fresh per packet.
- `TAG`: HMAC-SHA512 over `HEADER | u64(len(AAD)) | AAD | u64(len(CT)) | CT`, 64 bytes.

Key derivation (v2).

```
PRK     = HKDF-Extract-SHA512(salt = SALT, ikm)
info    = LP("MYSX-256/512 v2") | CAL | KDF | LP(context) | LP(label)
encKey  = HKDF-Expand(PRK, info(label="enc"), 64)
macKey  = HKDF-Expand(PRK, info(label="mac"), 64)
```

where `LP(x) = u16_be(len(x)) || x`.

v2 API. `v2.seal(plaintext, creds, opts)`, `v2.open(packet, creds, opts)`, `v2.generateSecret()`, `v2.securityLevel(creds)`, `v2.setArgon2Provider(fn)`, `v2.argon2FromNoble(argon2idAsync)`, `v2.CAL`, `v2.KDF`, `v2.DEFAULTS`, `v2.LIMITS`. The `_salt` and `_nonce` options are internal (KAT only) and must not be used in production.

Cross-language validation (v0.2). The v2 packet format has been implemented independently in Python (`tests/gen_v2_kat.py`, stdlib only) and in JavaScript (`mysx.prod.js`). Six complete packets (header + CT + tag) are **byte-identical** between the two implementations across: CAL=0 and CAL=1; KDF=RAW (64-byte and 32-byte secrets), PBKDF2-SHA512, scrypt; empty, 1-byte, 100-byte, 33-byte and 70-byte plaintexts. A 58-test suite (`tests/mysx_v2.test.js`) verifies KAT regression, cross-language KAT, per-byte bit-flip rejection, AAD/context/secret binding, KDF bounds, anti-downgrade on the KDF byte, password + keyfile separation, Argon2id RFC 9106 vector, and v1/v2 packet separation. All 58 tests pass.

2.10. Key derivation (v1, deprecated)

HKDF-SHA256 with 32-byte MAC key (or PBKDF2-SHA256 for passwords). The HKDF-SHA256 bottleneck caps the effective quantum security at `2^128`. Do not use v1 for new data.

2.11. Test vectors

- **`KAT.json` (v0.1):** block, AEAD, KDF vectors for the consecutive calendar. Public digest: `5c9ca63d19e0f385eb0142bbe0337771d32a25267469bc82c9cd8f0bed547eee`. Internal fast-regression digest `KNOWN_KAT_DIGEST = 3a8084d0b7df7eb691cfb54494410a52226`
(SHA-256 over 16 blocks with `key[i] = i`).
- **`KAT_v2.json` (v0.2):** `block_v02` (8 blocks with CAL=1), `v2_raw`, `v2_password_argon2/scrypt/p`
`v2_kdf`, `v2_parse`, `security_level`, `block_digest_cal0 == KNOWN_KAT_DIGEST`,
`block_digest_cal1 = d1b3e268af1c7422e38c8af6db1cc2be6c01b7c335423a7e5cd53a578171444`
- **`tests/_v2_kat_cross.json`:** 6 cross-language v2 packets (Python reference → JS must reproduce byte-identical).

A third-party implementation should target the public v0.1 digest, `block_digest_cal1` for v0.2, and the 6 cross-language v2 packets.

3. Differential cryptanalysis

3.1. Model

Tool: `misc/mysx_milp2.py` (PuLP + CBC/HiGHS, with `--r0` rotation offset). AND model: `a = Δx ∨ Δy`, counting cost-1 whenever at least one input is active. `gapRel = 0.2`, `timeLimit = 300–900` return feasible upper bounds (not certified optima). The model captures single trails, not differentials.

3.2. Observed upper bounds

At r0=0, v0.1 (input w2b30):

Rounds	Weight (upper bound)
5	4
6	6
7	10
8	18
9	26
10	42
11	58
12	68
13	88
14	144

At r0=0, v0.2 (input w1b31):

Rounds	Weight (r0=0)	Weight (r0=10)	Weight (r0=20)
10	59	92	76
11	62	92	76
12	98	129	86
13	120	139	104
14	129	164	132

Full run R=6..14 × r0=0,10,20: 27/27 solves.

Offset sweep, R=14, v0.1 (MILP, single-bit input w1b31): r0=10 → weight **2**, other offsets 50–266. Worst observed r0=30 → 266. Paper's beam values are tighter: 121 at r0=1 vs MILP 180 at r0=1, so the MILP overestimates by ~1.5–2× on these offsets.

Offset sweep, R=14, v0.2 (MILP, single-bit input w1b31): best **r0=29 → weight 94**, other offsets 94–243. Two offsets `r0=19`, `r0=27` timed out (`None`). The paper's differential minimum for v0.2 is therefore an upper bound of 94; the true minimum is likely 40–90 (MILP overestimates ~2×).

Comparison. v0.1 has a catastrophic offset (r0=10, weight 2 at R=14, 4 chosen pairs). v0.2 eliminates it: the best MILP weight across 31 offsets is 94 (upper bound). This is the main differential improvement of v0.2.

3.3. Direct verification of the r=4 trail

Trail: Δin = {w0b0}, MILP weight 8, Δout = {w0b4, w1b10, w1b16, w1b30, w6b4}. Verified with 8 fixed keys × 2^17 pairs, 128-round cipher with final whitening disabled.

Observed hit rate 7.802×10^-3 (8181/1048576), expected from MILP 2^-8 = 3.906×10^-3. Ratio **2.00×**. Consistent with differential clustering or a one-bit overestimate of the cost by the AND model.

v0.2 verification (informal). `diff_r4_v1` on v0.2 (with `MYSX_ROT_CAL` set) reports trail match 12/2000 (0.60%) at R=4 with average output difference weight 12.87 bits — the specific r=4 trail is not directly transferable to v0.2 (the schedule change perturbs the alignment at R=4). No new trail at R=4 has been identified for v0.2.

3.4. Differential growth at r0=10, v0.1

Beam search: R=14 → 2, 15 → 4, 16 → 6, 17 → 10, 18 → 15, 19 → 22, 20 → 34, 21 → 49, 22 → 78, 23 → 100, 24 → 133.

Verified with MILP (r0=10, input w1b31, CBC gapRel=0.2, timeLimit=180s): R=14 → 2, 15 → 4, 16 → 6, 17 → 10, 18 → 16, 19 → 24, 20 → 38, 21 → 56, 22 → 79, 23 → 107, 24 → 131.

Discrepancy: the paper's beam values differ from the MILP by −2 to +7 in the range R=18-24. Both are upper bounds; neither is a minimum. The MILP is (as expected)

slightly above the beam at every round; the paper's beam values are the tighter measurements for these offsets. The 128-bit threshold is reached at R ≈ 24 for both.

3.5. Differential weight-0 chains

A round has weight 0 iff the difference does not touch words 4..7 of the state; the round is a pure bit permutation on the left half.

v0.1 free chain table (verified, exact):

r0	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21–26	27	28	29	30
max	3	3	3	3	3	3	3	4	4	5	**13**	12	11	10	9	8	7	6	5	5	5	4	3	3	3	4

Maximum exact: By exhaustive linear algebra over the whole left-half difference space, no weight-0 chain longer than 13 rounds exists at any offset (v0.1).

Verified chains, v0.1 (fixed keys, 2^15 pairs per key × 8 keys, full 128-round cipher, final whitening disabled):

Offset	Δin	Δout	Rounds	Propagation
r0=10	w1b31	w4b12	13	**100.0000%**
r0=15	w2b30	w4b5	6	**100.0000%**

The 13-round chain at r0=10 covers rounds 10–22, 41–53, 72–84, 103–115 (period 31). It is the same chain at every occurrence.

v0.2 free chain table (verified, exact): maximum **5 rounds** at several offsets (r0 = 3, 4, 13, 16, 17, 20, 24, 27, 28); 4 rounds at most other offsets; 3 rounds at the remaining offsets. Full 31-offset table in `diff_free_chain_v02` output. No chain of length 6+ exists in v0.2.

Comparison. v0.1: 13-round exact chain. v0.2: 5-round exact chain. The reduction is 8 rounds, consistent with the exact DMITM reduction (17→9).

The catastrophic v0.1 offset. r0=10 in v0.1 gives both the 13-round weight-0 chain and the weight-2 trail at R=14 (§3.2). The two are the same alignment; eliminating one eliminates the other.

4. Linear cryptanalysis

4.1. Overview

The exact LAT of F (closed form) is used in a bit-exact mask propagation. Single-trail weights and hull sums are computed. A beam search over the exact model, both directions, up to 128 rounds, gives achievable single-trail weights (upper bounds on the best weight). Empirical verification is done with 2^22 plaintexts and fixed keys.

4.2. Exact LAT of F

For output mask b and input mask g:

```
b.F(x) ⊕ g.x  =  Q_b(x) ⊕ l.x,
l = rotr(b,2) ⊕ g,
Q_b(x) = ⊕_j b_{j+1} x_j x_{j−7}
```

Q_b is a quadratic form whose graph G_b is a subgraph of the 32-cycle j ↔ j−7. With A_b the GF(2) adjacency matrix:

- corr(b,g) ≠ 0 iff l = A_b y for some y. There are exactly 4^{h(b)} valid g, where h(b) = rank(A_b)/2.
- For each valid g, corr(b,g) = (−1)^{Q_b(y)} · c_0(b) with |c_0(b)| = 2^{−h(b)}.
- h(b) = maximum matching = minimum vertex cover of G_b. Exception b = 0xFFFFFFFF (full cycle): rank = 30, h = 15.
- Parseval: Σ_g corr(b,g)^2 = 1.

Exact and machine-checked against brute-force LAT of F for random b of Hamming weight ≤ 8.

Cost data.

- Single-bit b: h = 1, four valid g, |corr| = 1/2.
- Two-bit b (496 masks): histogram {2: 464, 1: 32}.
- Uniform random 32-bit b (16384 samples): histogram of h peaks at h = 11 (4130 samples).

4.3. Mask propagation

Forward step for round r with input mask m on s[r]:

```
b_p = m[p] (only words 0..3 decide cost)
g_p ∈ valid_gammas(b_p)  (4^{h(b_p)} choices, |corr|=2^{-h(b_p)})
n[p+4] = m[p]; n[p] = m[p+4] ⊕ g_p
m_next = lin_fwd(n, r)  (byte transposition, then rotl by ROT_CAL[r mod 31])
w_r = Σ_p h(m[p])
sign = (Π_p sign_p) · (−1)^{m · rk[r]}
```

Backward step: n = lin_inv(m_next, r); b_p = n[p+4], m[p] = n[p+4], m[p+4] = n[p] ⊕ g_p.

The selftest verifies that every forward edge is a backward edge with the same cost (325 edges checked).

4.4. Free-chain structure

A round has weight 0 iff words 0..3 of its input mask are zero; the round acts as a bit permutation and the tracked bit never meets F.

v0.1: 9-round weight-0 chains at windows r = 12 and r = 27 (mod 31).

v0.2: the same table (via `mysx_linear_beam.py reproduce`) is printed by `diff_free_chain_v02` and shows no weight-0 chains longer than ~9 rounds; the R=9 minimum for v0.2 forward is weight 1 (at offset 12), so the maximum weight-0 chain in the linear model is 8 rounds for v0.2 (consistent with the 5-round differential chain plus 3 free rounds that do not touch F).

4.5. Verified |corr| = 1 chains (v0.1)

9-round chains at r0=12 and r0=27 verified against the real cipher (fixed keys, 2^21 plaintexts, reduced rounds, final whitening disabled):

window	rounds	start bit	end bit	seed	predicted sign	empirical corr	σ
12	9	s[12].174	s[21].79	1	+1	+1.00000	6.9e-04
12	9	s[12].174	s[21].79	2	−1	−1.00000	6.9e-04
12	9	s[12].174	s[21].79	3	+1	+1.00000	6.9e-04
27	9	s[27].146	s[36].55	1	+1	+1.00000	6.9e-04
27	9	s[27].146	s[36].55	2	+1	+1.00000	6.9e-04
27	9	s[27].146	s[36].55	3	+1	+1.00000	6.9e-04

The magnitude is 1 for every key. Only the sign depends on the key.

4.6. Exact signed hulls (v0.1)

For trails at r0=12, R=10,11,12:

window	R	best-trail w	hull masks kept	predicted corr	empirical corr	σ	z
12	10	1	4	+0.500000	+0.499997	4.9e-04	−0.01
12	11	2	40	−0.250000	−0.250008	4.9e-04	−0.02
12	12	3	3152	−0.125000	−0.124813	4.9e-04	+0.38

Three independent predictions, three matches, |z| < 0.4.

v0.2 hulls (predictions only, no empirical). `lin_hull_v02` computes exact signed hulls for v0.2 with r0=12:

R	hull size	predicted corr
10	4104	−0.25
11	200	−0.125
12	10576	+0.031

Empirical verification (comparing predicted vs real cipher) for v0.2 is PENDING — the hulls are computed but not compared against `round_enc`.

4.7. What this establishes and what it does not (v0.1)

Established (machine-checked): exact LAT of F; exact mask propagation with key sign; |corr|=1 chains up to 9 rounds; exact signed hulls at r=10,11,12 matching the real cipher within |z|<0.4.

Not established: a lower bound on the best trail weight over 128 rounds; an exact 128-round hull; a hull-aware correlation for the §4.10 trails beyond R=14.

4.8. Previous incorrect models

Recorded for context: v0.0 draft reported three approximate linear MILP models, all failed empirical verification. They used per-cell AND activity (`act = α ∨ β ∨ γ`) as a proxy for correlation, which is not the correct structure of F.

4.9. Integral / division property

Standard bit-based division property: COPY, XOR with (1,1) invalid, AND. `--solver highs` for feasible runs.

At r0=0, v0.1:

Input	Active bits	Integral size	Rounds with balanced bits
bit 0 of word 0	1	2	10 (single-bit dependency, not integral)
word 0	32	2^32	6
words 0 and 4	64	2^64	5

At r0=15, v0.1: 10-round distinguisher for both patterns (thin tail: 1 balanced bit at R=10, 0 at R=11).

At r0=0, v0.2:

Input	R	Balanced bits
word 0	5	**9/256**
word 0	6	**4/256**

At r0=0, v0.1 (measured with `integral_v1_R6`): 4/256 at R=6, matching the paper. Full word0,word4 at R=5: 11/256.

Offset sweep for v0.2. Only 6 offsets measured (0, 5, 10, 15, 20, 27) at R=5-6 via `run_integral_sweep_key6.py`. The full 31-offset sweep is PENDING.

Not established for v0.2: integral distinguishers at r0=15 (the paper's peak for v0.1); full 31-offset sweep; whether v0.2 has a r0=15-like offset with a 10-round distinguisher; three-subset DP.

4.10. Beam search over the exact linear model

v0.1 results (verified, see §11 for reproduction): single-trail weights at R = 9, 10, 12, 14, 16, 18, 20, 22, 24, 32, 40, 48, 56, 64, 80, 96, 112, 128.

R	weight	offset	direction
9	0	4	backward
10	1	4	backward
12	2	20	backward
14	5	20	backward
16	12	20	backward
18	24	20	backward
20	38	20	backward
22	71	3	backward
24	88	3	backward
32	204	3	backward
40	361	20	backward
48	485	20	backward
56	**653**	20	backward
64	**815**	13	backward
80	1081	21	backward
96	1336	21	backward
112	1671	21	backward
128	2248	3	backward

Correction vs v0.1 paper. The paper's beam table lists R=56 = 630 at offset 27 forward and R=64 = 786 at offset 27 forward. The committed `beam_full.json` and `beam_bwd.json` give 653 at offset 20 backward and 815 at offset 13 backward respectively. The paper's values are not reproducible from the committed JSONs. The 16 of 18 checkpoints that match are asserted in `lin_beam_full_table_v1`; the 2 discrepancies are documented (not asserted).

v0.2 results (from `lin_beam_v02_R128` forward, all 31 offsets):

R	weight	offset	direction
9	1	12	forward
10	1	14	forward
12	3	14	forward
14	8	14	forward
16	18	14	forward
18	39	14	forward
20	71	14	forward
22	95	14	forward
24	124	14	forward
32	256	14	forward
40	396	14	forward

R	weight	offset	direction
48	541	14	forward
56	701	13	forward
64	839	13	forward
80	1127	7	forward
96	1439	7	forward
112	1790	7	forward
128	2294	0	forward

v0.2 backward beam to R=128 is PENDING.

Comparison at R=14: - v0.1 backward (offset 20): weight 5 - v0.2 forward (offset 14): weight 8 - v0.2 backward at R=14 (measured): weight 11 (offset 11) — see `lin_beam_v02_R14_bwd`.

The comparison v0.1 vs v0.2 at R=14 is: 5 (v0.1, backward) vs 8 (v0.2, forward) vs 11 (v0.2, backward). v0.2 is slightly better in forward, slightly worse in backward, than v0.1 at this round count. Neither is “much stronger”.

R=128 weight: v0.1 backward = 2248; v0.2 forward = 2294. Essentially equal (both irrelevant at 2^4496).

4.10.3. Hull check of trails

v0.1 (verified, offset 20 backward, R=14, w=5): isolated in its hull (hull_count returns 1 trail). Empirical correlation matches 2^-5 on the real cipher at every prefix. This is the verified distinguisher on a 16-round window (rounds 98–113) reported in v0.1 §4.10.3.

v0.2: no empirical hull check performed. `lin_hull_v02` computes exact signed hulls but does not compare with the real cipher. PENDING.

Cancelation (v0.1, verified). r0=23 forward cancels from R=8; r0=24 forward cancels from R=8. Single-trail weight is not universally a distinguisher length.

4.10.5. Exact hull at R=15–17 (v0.1)

The offset-20 backward trail at R=16 has exact hull |corr| = 2^-11 (two trails of weight 12 adding). The empirical measurement is +6.06×10^-4 ± 6.1×10^-5 against a prediction of 4.88×10^-4 (+1.9σ). At R=17, the best end pairs of offsets 20 backward and 23 forward cancel exactly within the enumerated budget, for both keys tested. R ≥ 18 is undecided.

4.11. Statistical randomness of the CTR keystream

v0.1: CTR keystream passed NIST SP 800-22 (100 sequences × 10^6 bits, fresh key and nonce per sequence, no failures). Full report in `LEGACY_V1/NIST_SP_800-22_report.txt`.

v0.2: PENDING. The `nist.yml` workflow fails at "Download and build NIST STS 2.1.2" (build error not yet diagnosed). The v0.2 block cipher uses the same core as v0.1 with a different calendar; the NIST battery is expected to pass by analogy, but has not been measured.

4.12. Impossible differentials

Weight-0 chains. Verified table in §3.5. v0.1: max 13 rounds at r0=10. v0.2: max 5 rounds at several offsets.

Exact DMITM.

Version	max weight-0 chain	exact DMITM
v0.1	13 rounds	**17 rounds**
v0.2	5 rounds	**9 rounds**

v0.1 verification: `dmitm_v1` confirms max=13, total=17 with 0 offsets where multi-bit improves over single-bit. The best witness is Δs[23]=w1b31 –13 rounds–> w4b12, Dout=w7b6 <–4 rounds– w3b16, incompatibility at s[23]. Verifiable in 0.3s.

v0.2 verification: `dmitm_v02` confirms max=5, total=9. The best witness is Δs[3]=w3b16 –4 rounds–> w6b18, Dout=w7b7 <–5 rounds– w0b30, incompatibility at s[3].

Truncated impossible (v0.1). Δs[44]=w2b29 –18 rounds–> s[62] <–21 rounds– Δs[83]=w7b24, conflict at **w6b16**. This is **39 rounds**. Verified with `dmitm_trunc_v3.py --cal k1`: premises 100% over 2000 pairs × 4 keys per side (0x00, 0xFF, and 2 random); propagation of the witness is exact and the conflict at w6b16 is confirmed.

Truncated impossible (v0.2). Δs[66]=w2b25 –8 rounds–> s[74] <–20 rounds– s[94]=w4b14, conflict at w5b13. This is **28 rounds**. Verified with `dmitm_trunc_v3.py --cal v02 --verify 500`: premises 2000/2000 per side against the real cipher.

Reduction. 39 → 28 (11 rounds). Consistent with the reduction in exact DMITM (17→9, −8) and weight-0 chain (13→5, −8). The truncated figure is the exact figure plus a short weighted prefix, so both shrank by approximately the same amount.

Not established. The search covers single-bit starting masks on all 31 rotation offsets. Multi-bit starts, weight-1 rounds, and narrower propagation are PENDING. The `--verify` check confirms the premises but is not a key-recovery cost. The truncated witness is a distinguisher, not an attack.

4.13. Rotation schedule selection

The 9-metric comparison (v0.1 vs v0.2, from `wf2_deep_summary.json`):

Metric	v0.1	v0.2	Result
dp5 (integral R=5)	21	**9**	improvement
dp6 (integral R=6)	4	4	tie

Metric	v0.1	v0.2	Result
lin_fwd10	1	1	tie
lin_bwd10	1	**3**	improvement
lin_fwd14	5	**8**	improvement
lin_bwd14	5	**11**	improvement
DMITM exact	17	**9**	improvement
Weight-0 chain	13	**5**	improvement
MILP R=5	0	**4**	improvement
Truncated	39	**28**	improvement (−11)

Directions. Lower is better for dp5, dp6, DMITM, chain, truncated. Higher is better for lin_fwd10/14, lin_bwd10/14, MILP.

6 improvements, 3 ties, 0 regressions.

A false positive documented. `rand_s9_33719` was top-ranked by R=4 screening (with dp4=42, milp5=10). At R=6 its integral is worse than v0.1 (dp6=12 vs 4) and its linear weight at R=14 is 13 vs 5. Screening at R≤4 is not sufficient.

Why Myszkowski is not the cause. A control with 40 random byte permutations and the same consecutive schedule gives mean exact DMITM 21.4 (range 10–34). MYSX v0.1 is at ~22nd percentile. Changing the consecutive schedule to `(k·r mod 31)+1` for k ∈ {2,3,5,7} drops the maximum chain to 6–9 rounds. The variable is the schedule step, not the permutation.

4.14. What is PENDING for v0.2

The following are measured for v0.1 and not re-measured for v0.2:

- **Differential:** full 31-offset sweep (only 6 offsets measured; best r0=29 weight 94; r0=19, r0=27 unresolved); beam differential (only MILP for v0.2); cluster analysis at v0.2 offsets.
- **Linear:** backward beam to R>14 and R=128; hull empirical verification.
- **Integral:** full 31-offset sweep outside r0=0; r0=15 peak; R=7–10 tail; full $2^{32/2}$64 integral verification.
- **Impossible:** multi-bit starts; weight-1 rounds; narrower propagation.
- **NIST:** full NIST SP 800-22 battery.
- **Related-key:** 22-test re-validation (key schedule unchanged; expected identical).
- **Simon normalized coverage comparison:** v0.1's comparison was against the consecutive schedule for MYSX; v0.2 has different figures.
- **Differential R=16, R=17 empirical trails** (`diff_r16_r17_v1`, pending external reviewer data).

5. Related-key cryptanalysis

Inherited from v0.1 (key schedule unchanged). 22 structural and empirical tests up to weight-6, no persistent Δrk[r]=0 for r≥1, no weak keys, no cycles, no fixed

points. Slow diffusion at r=1–4 (expected: one Feistel round per cipher iteration). Worst bit: bit 160 (word 5, bit 0), 1-bit difference at r=1; consistent across 5 base keys (structural, not noise). Beam search score 139 (ideal 128).

Limitations. Heuristic, not a proof. No targeted MILP/SAT related-key model. Adaptive related keys, biclique attacks, algebraic attacks on the key schedule not covered. Related-key security is unknown.

Round-key recovery. The key schedule is bijective; any 16 consecutive state words determine the master key. How many consecutive round keys a SAT solver needs to recover the master key (about three by counting) has not been measured.

6. Generic attacks

Attack	Complexity (uniformly random 512-bit secret)
Classical brute force	2^512 operations
Grover (quantum)	2^256 iterations
Block collisions	2^128 blocks

Note on Grover (v2 API). A uniformly random 64-byte secret processed through HKDF-SHA512 gives 2^256 Grover per target. `2^256 / sqrt(N)` if an attacker needs only one of N messages.

Note on the v1 API. HKDF-Extract with HMAC-SHA256 produces a 256-bit PRK. The effective quantum security is 2^128, not 2^256. Do not use v1 for new data.

Note on password paths. Security is limited by password entropy, not by the nominal key size. `v2.securityLevel` returns null for classicalBits/groverBits when the password is not accompanied by a ≥64-byte keyfile. A keyfile is strongly recommended.

7. Simon cryptanalysis: what transfers

MYSX uses Simon's F. It is not Simon. Differences: block size, branch organization (4 parallel Feistel cells, Simon uses 1), Myszkowski transposition (Simon has none), round-dependent word rotations (Simon has none), non-linear ARX key schedule (Simon's is linear), 128 rounds vs 68.

What transfers. The exact LAT of F (§4.2) — its closed form, its validity condition, its sign structure — is a property of the function and applies to MYSX regardless of surrounding structure.

What does not transfer. Key recovery from Simon's linear key schedule (MYSX is non-linear). Linear hull attacks structured around Simon's Feistel branch. Integral distinguishers relying on Simon's 2-branch Feistel structure and linear key schedule.

Published Simon results (context only): Chen-Wang FSE 2016 (49-round linear-hull key-recovery on Simon128/128); Shi et al. ePrint 2014/973 (34-round linear characteristic, corr 2^-61 for Simon128); Biryukov-Roy-Velichkov FSE 2014 (differential); Todo-Morii FSE 2016 (14-round integral on Simon32, 15 with three subsets); Xiang-Zhang-Bao-Lin ASIACRYPT 2016 (MILP division property).

8. Analysis limitations

Differential: single-trail upper bounds; two independent deterministic chains verified empirically (13 rounds at r0=10, 6 rounds at r0=15, both v0.1); no lower bounds; no full clustering analysis at r0≠0; the exact maximum of weight-0 chains (13 for v0.1, 5 for v0.2) is proven by linear algebra; the non-zero-weight tables are upper bounds.

Linear: exact model of F; exact signed hulls at r=10,11,12 (v0.1) matching the real cipher within |z|<0.4; beam search over the model in both directions (v0.1) and forward (v0.2), 31 offsets, up to 128 rounds. v0.2 backward beam to R=128 is PENDING. No lower bound over 128 rounds. No hull-aware bound beyond R=17 for v0.1. v0.2 hull empirical verification is PENDING.

Integral: v0.1 has 6-round (2^32) / 5-round (2^64) at r0=0, 10-round at r0=15; v0.2 has 6-round / 5-round at r0=0, 6-offset sweep at R=5-6. Full 31-offset sweep for v0.2 and r0=15 peak for v0.2 are PENDING.

Impossible: v0.1 exact = 17, truncated = 39 (verified); v0.2 exact = 9, truncated = 28 (verified). Multi-bit starts and weight-1 rounds PENDING.

Related-key: heuristic (v0.1 inherited).

NIST: v0.1 pass; v0.2 PENDING (workflow build error).

Other: boomerang, algebraic, side-channel not analyzed. No external audit.

9. Engineering notes

9.1. API surface

Public (v2): `v2.seal/open`, `v2.generateSecret`, `v2.securityLevel`, `v2.setArgon2Provider`, `v2.argon2FromNoble`, `v2.CAL`, `v2.KDF`, `v2.DEFAULTS`, `v2.LIMITS`.

Public (v1, deprecated): `aeadEncrypt`, `aeadDecrypt`, `deriveAeadKeys`, `deriveAeadKeysFromPassword`, `NonceGuard`, `CONFIG`.

Test API: disabled by default. Enabled only with `MYSX_ENABLE_TEST=1`.

9.2. Migration v1 → v2

- v1 packets (0x01…) decryptable with `aeadDecrypt`.
- v2 packets (0x02…) created by `v2.seal`, not decryptable by `aeadDecrypt`.
- Do not use `aeadEncrypt` for new data (HKDF-SHA256 → 2^128).
- Do not pass `_salt`/`_nonce` to `v2.seal` in production (internal parameters for KAT generation).

9.3. NonceGuard (v1)

Best-effort defense against nonce reuse in v1. The v2 API does not need NonceGuard: every message uses a fresh 32-byte salt and 16-byte nonce.

9.4. Type validation

All public functions validate byte inputs are Uint8Array of expected length.

9.5. KAT digests

- `KAT.json` (v0.1): digest `5c9ca63d19e0f385eb0142bbe0337771d32a25267469bc82c9cd8f0bed547e`
- `KAT_v2.json` (v0.2): `block_digest_cal0 == KNOWN_KAT_DIGEST`; `block_digest_cal1 = d1b3e268....`
- `tests/_v2_kat_cross.json`: 6 cross-language v2 packets.
- `KNOWN_KAT_DIGEST = 3a8084d0b7df7eb691cfb54494410a5222652c219a336ee58bde48a64524a8b3`

9.6. Harness validation

- Linear sampling harness: fixed key gives |corr| = 0.4993; random key gives ~0.0022. Confirms the harness is correct with fixed keys.
- mini-MYSX w=32 reproduces mysx.prod.js exactly. Keyword/w compatibility: `MYSZKOWS` at w=32, `MYSZ` at w=8 (non-degenerate).
- Beam-search tool `selftest` passes in both directions (LAT of F, one round, 9-round chains, forward/backward duality, backward R=14 trail).
- Differential verification scripts check the deterministic trails of §3.5 at 100% propagation.
- Division-property harness validated against R=10 single-bit dependency (2 balanced bits confirmed at 20,000 samples).

10. Verdict

MYSX-256/512 v0.2-draft is an experimental proposal, not a secure cipher.

Rotation schedule. `rand_s9_45430` replaces the consecutive schedule `(r mod 31)+1`. On the metrics measured, the new schedule improves 6, ties 3, and regresses 0. Exact DMITM: 17 → 9. Weight-0 chain: 13 → 5. Truncated impossible: 39 → 28 (verified 2000/2000 against the real cipher). Integral balanced bits at R=5: 21 → 9. MILP differential R=5: 0 → 4. **The v0.1 catastrophic offset r0=10 (weight-2 trail at R=14, 4 chosen pairs to distinguish 14 rounds) is eliminated in v0.2 (best MILP weight at R=14 is 94; two offsets unresolved).**

AEAD v2. HKDF-SHA512 with a 64-byte secret (2^256 Grover); 512-bit MAC keys; HMAC-SHA512; one key per message; authenticated header with rotation calendar; Argon2id for password paths. Cross-language KAT: 6/6 packets byte-identical to an independent Python reference. 58 security tests pass.

PENDING for v0.2. Differential beam (MILP only); full 31-offset differential sweep (6 offsets measured, 2 unresolved); linear backward beam to R=128; hull empirical verification; full 31-offset integral sweep (6 offsets measured); NIST SP 800-22. The v0.2 analysis is incomplete by design.

No lower bound over 128 rounds is established. No key-recovery attack has been attempted. No external audit. The truncated impossible for v0.2 is 28 rounds, not 39. The related-key security is unknown.

Do not use MYSX to protect real data. For production: AES-256-GCM or XChaCha20-Poly1305.

11. Reproducibility

All measured claims of §2–§5 are reproducible via `./run_all_gh_actions.sh --no-nist` in the repository. 49 tests pass (0 fail, 2 skip: NIST and `diff_r16_r17_v1`).

Files

File	Purpose
`mysx.prod.js`	Reference implementation (v1 + v2)
`mysx.test.js`	17 functional tests (v1)
`KAT.json` (in `LEGACY_V1/`)	v0.1 test vectors
`KAT_v2.json`	v0.2 test vectors
`calendar/CALENDAR_v02.txt`	`MYSX_ROT_CAL=22,5,8,...`
`calendar/wf2_deep_summary.json`	8 candidates × 9 metrics (§4.13)
`cryptanalysis/DMITM/dmitm_verify.py`	Exact DMITM search
`cryptanalysis/DMITM/dmitm_trunc_v3.py`	Truncated impossible differentials (§4.12)
`cryptanalysis/DMITM/search_mysx_deep.py`	1M random schedule search
`cryptanalysis/models/linear/mysx_linear_model.py`	Exact linear model of F + round
`cryptanalysis/models/linear/LINEAR_MODEL.md`	Model documentation
`cryptanalysis/models/linear/linear_model_data.json`	Data for §4.2–§4.6
`misc/mysx_linear_beam.py`	Beam search over the exact linear model
`misc/beam_full.json`, `misc/beam_bwd.json`	v0.1 beam outputs
`misc/beam_paths_bwd20.json`, `misc/beam_paths_fwd23.json`	Hull inputs (§4.10.5)
`misc/verify_beam_trail.py`, `misc/hull_count.py`, `misc/hull_bidir.py`	Hull checks
`misc/mysx_milp2.py`	Differential MILP
`misc/mysx_division_property_v2.py`	Division property
`tests/gen_v2_kat.py`	Independent Python reference for v2 packets
`tests/mysx_v2.test.js`	58 security tests for v2 API
`tests/manifest.json`	Test manifest (53 tests, 51 ready + 2 skip)
`run_all_gh_actions.sh`, `run_all_local.sh`	Orchestrators
`.github/workflows/paper.yml`	CI matrix for non-NIST tests
`.github/workflows/nist.yml`	CI matrix for NIST (build broken)

Reproduction commands

v0.1 claims:

```
# KAT
python3 -c "import json; d=json.load(open('LEGACY_V1/KAT.json')); \
  assert d['digest']=='5c9ca63d19e0f385eb0142bbe0337771d32a25267469bc82c9cd8f0bed547eee

# Free chain table (r0=10 -> 13, r0=15 -> 8, r0=0 -> 3, r0=27 -> 3)
python3 misc/diff_free_chain_table.py

# 13-round chain verification
python3 misc/verify_diff_13r.py    # 100.0000%

# 6-round chain verification
python3 misc/verify_diff_trail.py  # 100.0000%

# MILP differential (R=5 -> 4, R=14 -> 144)
python3 misc/mysx_milp2.py --rounds 5-14 --input-bit 94 --r0 0 --solver cbc

# Beam reproduce (paper §4.5)
python3 misc/mysx_linear_beam.py reproduce

# Exact hulls (|z| < 0.4)
python3 tests/run_hull_v1.py

# Exact DMITM (17 rounds)
python3 cryptanalysis/DMITM/dmitm_verify.py exact

# Truncated impossible (39 rounds)
python3 cryptanalysis/DMITM/dmitm_trunc_v3.py --cal k1 --max_fwd 24 --max_bwd 24

# Beam to R=128 backward (2248)
python3 misc/mysx_linear_beam.py run --offsets 0-30 --rounds 128 --beam 2000 --jobs 4 -

# Hull check (isolated at R=14, cancels at r0=23 forward)
python3 tests/run_hull_check_v1.py
```

v0.2 claims:

```
export MYSX_ROT_CAL=22,5,8,7,17,10,30,1,11,26,6,3,12,23,25,13,28,20,21,19,24,2,18,9,4,1

# Exact DMITM (9 rounds)
python3 cryptanalysis/DMITM/dmitm_verify.py exact

# Free chain table (max 5 rounds, no 13)
python3 misc/diff_free_chain_table.py

# Truncated impossible (28 rounds with VERIFY)
python3 cryptanalysis/DMITM/dmitm_trunc_v3.py --cal v02 --max_fwd 24 --max_bwd 24 --ver

# Beam R=10 forward (1) and backward (3)
python3 misc/mysx_linear_beam.py run --offsets 0-30 --rounds 10 --beam 2000 --jobs 1
```

```
python3 misc/mysx_linear_beam.py run --offsets 0-30 --rounds 10 --beam 2000 --jobs 1 --

# Beam R=14 forward (8) and backward (11)
python3 misc/mysx_linear_beam.py run --offsets 0-30 --rounds 14 --beam 2000 --jobs 1
python3 misc/mysx_linear_beam.py run --offsets 0-30 --rounds 14 --beam 2000 --jobs 1 --

# Beam R=128 forward (2294)
python3 misc/mysx_linear_beam.py run --offsets 0-30 --rounds 128 --beam 2000 --jobs 2 -

# Integral R=5 (9/256), R=6 (4/256)
python3 misc/mysx_division_property_v2.py --rounds 5 --active word0 --r0 0 --solver hig
python3 misc/mysx_division_property_v2.py --rounds 6 --active word0 --r0 0 --solver hig

# MILP R=5 (4, min over r0 in {0,10,20})
python3 tests/run_diff_milp_v02_R5.py

# Hull predictions (r=10,11,12)
python3 tests/run_hull_v02.py

# Cross-language v2 KAT + security suite
python3 tests/gen_v2_kat.py
MYSX_ENABLE_TEST=1 node tests/mysx_v2.test.js
```

12. Final warning

MYSX has not been externally audited. This paper is the author's self-analysis, not a certification. No result here replaces an independent cryptographic review.

Do not use MYSX to protect real data. For production: AES-256-GCM or XChaCha20-Poly1305.